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A proper choice of route significantly reduces air pollution exposure — A study on bicycle and bus trips in urban streets

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ABSTRACT

A proper selection of route through the urban area may significantly reduce the air pollution exposure. This is the main conclusion from the presented study. Air pollution exposure is determined for two selected cohorts along the route going from home to working place, and back from working place to home. Exposure is determined with a street pollution model for three scenarios: bicycling along the shortest possible route, bicycling along the low exposure route along less trafficked streets, and finally taking the shortest trip using public transport. Furthermore, calculations are performed for the cases the trip takes place inside as well as outside the traffic rush hours. The results show that the accumulated air pollution exposure for the low exposure route is between 10% and 30% lower for the primary pollutants (NO_x and CO). However, the difference is insignificant and in some cases even negative for the secondary pollutants (NO_2 and $\text{PM}_{10}/\text{PM}_{2.5}$). Considering only the contribution from traffic in the travelled streets, the accumulated air pollution exposure is between 54% and 67% lower for the low exposure route. The bus is generally following highly trafficked streets, and the accumulated exposure along the bus route is therefore between 79% and 115% higher than the high exposure bicycle route (the short bicycle route). Travelling outside the rush hour time periods reduces the accumulated exposure between 10% and 30% for the primary pollutants, and between 5% and 20% for the secondary pollutants. The study indicates that a web based route planner for selecting the low exposure route through the city might be a good service for the public. In addition the public may be advised to travel outside rush hour time periods.

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1. Introduction

A large number of studies have shown strong associations between ambient air pollution levels and adverse health effects; see e.g. the review by (Brunekreef and Holgate, 2002). Studies of long-term exposure to air pollutants have suggested an increased risk of chronic respiratory illness (Folinsbee, 1993), cardiopulmonary mortality (Hoek et al., 2002) and of developing various types of cancer (Pope et al., 2002), whereas higher prevalence of bronchitis (Karakatsani et al., 2002), acute cardiovascular disease (Urch et al., 2005), asthma and other symptoms (Sunyer et al., 1997) have been associated with

short-term exposure to air pollution during periods with enhanced concentration levels.

Inside the urban area, the dispersion of pollutants emitted from traffic is strongly suppressed by the presence of building obstacles. The traffic density is at the same time often substantial in the urban areas, and in cities like Copenhagen it has even been increasing in recent years. Traffic is thus a source to significant air pollution levels in urban areas in general, but especially inside the busiest streets. People living and/or working in the urban areas are therefore exposed to high levels of air pollution, especially when they are travelling in the trafficked streets.

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Most of the reported associations between exposure to air pollutants and health effect are the results of time-series analyses based on data for mortality and morbidity rates in combination with demographic data; e.g. information about population densities, addresses etc. Exposure estimate are here usually obtained from associating the demographic data with pollution measurements obtained from a limited number of fixed site monitoring stations distributed over the urban area. This approach is questionable when monitoring data from street stations are used in the exposure assessments, since these stations represent very local conditions. Local conditions that to a large extent are governed by the traffic in the specific street in combination with parameters that are determining the local dispersion conditions; for the latter, especially the street width and the height of buildings surrounding the street are important. For short time averages of air pollution levels e.g. hourly mean values, there may be an order of magnitude in concentration difference just between the pavements on the two sides of the street (Berkowicz et al., 1996). There are, however, examples of studies, also for Copenhagen, where exposure estimates have been obtained from personal monitors, and where such directly measured exposures have been linked to biomarkers for adverse health effects (as well as biomarkers for exposure) (Sorensen et al., 2003; Sorensen et al., 2005). These studies have shown that oxidative stress-induced DNA damage in blood is linked to exposure to PM_{2.5} and black carbon. In exposure studies of humans, systemic effects have been found for traffic-related particles in terms of decreased endothelial function, which is important for the development of cardiovascular disease as well as oxidative stress-induced DNA damage (Vinzents et al., 2005; Mills et al., 2005). These and other studies of short-term air pollution exposure are conducted on data for the human air pollution exposure over 24 hour or even 48 hour periods. However, little is known about the necessary duration of enhanced air pollution exposure in order to increase the risk of health effects. One exception is a German study where electrocardiogram recordings were compared to continuous recordings of particle and gaseous air pollution that has shown that elevated concentrations of fine and ultrafine particles increased the risk of arrhythmia in men with coronary heart disease (Berger et al., 2006).

Although air pollution levels in Denmark are generally low compared with what is observed in many other larger cities in Europe and elsewhere, the measured NO₂ and PM₁₀ concentrations in busy streets of Copenhagen have been reported in some cases to exceed the EU Guidelines (Kemp et al., 2006). The population in most countries is thus exposed to high concentrations when travelling in the busiest urban streets. The average air pollution exposure during these travels depends among other things on the means of transport. A Dutch study has thus shown that car drivers in urban streets may be exposed to higher pollutant levels than bicyclists travelling in the same streets. This is due to exhaust gases from cars in front that in some cases may be entering the vehicle directly through the ventilation system (vanWijnen et al., 1995). This enhanced air pollution exposure is, however, somewhat compensated by the higher respiration rate of bicyclist when comparing the received air pollution dose for two groups. Somewhat similar results were found in a

Danish study also concerning car drivers and bicyclists (Rank et al., 2001).

The dispersion conditions inside an urban street are highly affected by the configuration of the specific street. The street configuration can here be defined by a number of characteristic parameters: the street width, the orientation of the street, the building height along the street, eventual openings in the building facade along the street, and the length of the considered street section (Berkowicz et al., 1996). Naturally, these parameters are together with the distribution and composition of traffic, varying considerably from one street to another. Large variations may often be observed, even from one section of the street to next. These differences in street configuration and traffic lead to significant differences in pollution levels, even between points separated by very short distances.

In Denmark, the bicycle is a common mean of transport over short distances e.g. in urban areas. When the air pollution levels vary significantly between streets and even between street sections, there must be a significant difference in air pollution exposure depending on the choice of route through the city. This consideration is the background for the present study, where we investigate the influence of choosing high and low exposure routes in urban streets when commuting between home and working place. The main questions that we address in this paper are therefore:

- Is it possible to significantly reduce air pollution exposure during a daily bicycle trip between home and work by taking a low exposure route through a city, as an alternative to taking the shortest possible route?
- Can the air pollution exposure be significantly reduced by travelling outside rush hours time periods compared to travelling during the rush hour?
- To what extent is the exposure during travels in urban areas dominated by the urban background contribution, and the contribution from local traffic emissions, respectively?
- How is the air pollution exposure for the same travel from home to working place when you take the bus compared to taking the bicycle?
- Is there a potential for developing a green route planner to help the public in choosing the route through the city with the lowest air pollution exposure?

2. Model tools and input data

The Operational Street Pollution Model (OSPM) (Berkowicz, 2000) is applied for calculating air pollution concentrations in streets along the routes in the Greater Copenhagen Area. The OSPM is able to handle various street configurations, and the model includes emission factors and standard traffic distributions for different street categories in Denmark (Berkowicz et al., 2006). The presented calculations are based on observed urban background concentrations and meteorology obtained from a monitoring station placed on the roof top of the university building of the H.C. Ørsted Institute in central Copenhagen. The OSPM handles traditional traffic pollutants and contains in addition a chemical module for computing

NO₂ concentrations in the street (Palmgren et al., 1996). In the performed calculations, all traffic data are representing the year 2005. The measured urban background levels and the measured meteorological data are representing the year 2003.

The high quality of the OSPM is internationally accepted, and the model has been applied for modelling street pollution in a large number of mainly European cities (Aquilina and Micallef, 2004; Gokhale et al., 2005; Ketzel et al., 2000; Kukkonen et al., 2000; Mensink et al., 2006; Vardoulakis et al., 2002; Fu et al., 2000; Vardoulakis et al., 2005, 2007). The OSPM calculates pollutant concentrations on hourly basis at kerbside. The OSPM is a fast tool and considered to represent state-of-the-art in simple parameterized models. The model has in previous studies been shown to reproduce well the observed air pollution levels in urban streets. The best model performance has been obtained for the traditional gaseous traffic pollutants CO, NO_x and NO₂ and for typical street canyons. Particle number concentrations have been modelled in one study and showed a good agreement with measurements (Ketzel, 2004). Particle mass has been modelled in a couple of studies and these studies have shown a somewhat weaker model performance. The reason is that only a fraction of the particle mass is emitted directly with the vehicle exhaust, whereas a large part of the particle mass is related to resuspension of dust and wear of road material, tires, breaks etc. This makes it difficult to obtain a proper estimation of the emission of particle mass. Most of the OSPM validation studies have concerned monitoring sites in street canyons, for which also the best model performance is obtained. In addition to the referred validation studies, the model performance is here illustrated for two street pollution monitoring stations in central Copenhagen. Figs. 1 and 2 show comparisons between measured and modelled concentrations of NO_x and PM_{2.5}/PM₁₀ for Jagtvej and H.C. Andersen's Boulevard, respectively. The plots show diurnal mean values and average diurnal profiles, since focus in this context is on the models ability to predict long-term averages for use in exposure assessment. PM is measured with TEOM instruments that give lower values than the reference method due to evaporation of volatile material during the collection at temperature of 50 °C. The best model performance is obtained for Jagtvej which is a regular street canyon with buildings on both sides of the street. The H.C. Andersen's Boulevard has buildings on one side and is open to the opposite side. The more complex street configuration of the H.C. Andersen's Boulevard explains parts of the difference in model performance for the two streets. Similar results as for NO_x are found for the NO₂ and CO concentrations (not shown here). The NO_x and CO levels are in general reproduced better than the levels of PM₁₀ and PM_{2.5}. The gaseous pollutants NO_x and CO are emitted only with the vehicle exhaust, whereas PM is emitted also from resuspension of dust and wear of tires, breaks and road material (Ketzel et al., 2003, in press). The emission factors are therefore in general better defined for CO and NO_x than for PM₁₀ and PM_{2.5}. Moreover, the ratio between background concentration and street contribution is much higher for PM than for NO_x. The uncertainty in the description of the PM emission processes introduces a larger uncertainty for the estimation of the street increment for PM, since a smaller difference is calculated and this leads to the large scatter observed in Figs. 1D and 2D. The average diurnal profile

for the pollutant concentrations in the street (plots E and F in Figs. 1 and 2) is generally very well reproduced in the model calculations. The modelled street pollution concentration (plots A and B) includes the measured urban background level. A comparison based on the street contribution alone (plots C and D) is thus a more direct test of the model performance.

The OSPM has been applied previously for assessment of human exposure to air pollution. In the Danish children cancer project (Raaschou-Nielsen et al., 2001), the OSPM model was used for assessment of air pollution exposure at the residence addresses. The front door pollution level was used as proxy for the exposure of the children, and input data for the OSPM calculations were obtained from questionnaire filled in by the local authorities. The model performance was tested in a pilot study where passive samplers were applied to measure NO₂ and benzene levels at 200 addresses; 100 in the centre of Copenhagen city and 100 in the rural suburbs. The OSPM was shown to reproduce well the observed levels at the addresses (Raaschou-Nielsen et al., 2000). In the Danish bus driver study, the OSPM was applied for assessment of air pollution exposure of bus drivers in Copenhagen city (Wilhardt et al., 1996; Hertel et al., 2001a,b). Bus schedules and diaries were used to obtain information about when the bus was present in the various streets along the route. Input data for the OSPM calculations were obtained from the local authorities and from on-location observations. Passive samplers were applied to monitor average NO₂ pollution levels over an entire working day (8 h) inside and outside the bus during a three-week campaign. OSPM was applied for calculating air pollution levels in 22 different street sections along the route. A programme was constructed to handle the OSPM results and compute the average exposure during an entire working day of the bus driver. The results showed that pollutant levels inside and outside the bus were almost identical. Furthermore the OSPM calculations reproduced very well the observed levels.

The Danish AirGIS system (Jensen et al., 2001) consists of a GIS platform coupled to Danish registry data and digital maps (see also <http://AirGIS.dmu.dk>). AirGIS includes routines for generation of input data for personal exposure calculations on address level in connection with epidemiological studies. In those studies large numbers of exposure calculations at address level (>200,000 addresses) are needed for a series of often very large cohorts in Denmark. In the presented work, AirGIS is used for the generation of the street configuration data needed for performing the calculations with the OSPM.

3. Study design and methodology

The analysis is carried out for two selected cohorts each consisting of 25 persons. The addresses were selected from a random subset of Danish residential addresses, and any possible association to real individuals are unintended and purely coincidental. The exposure is determined for working days along the route going from home to working place, and back from working place to home. Calculations are performed for three different scenarios: bicycle along the shortest route; bicycle along the low exposure route following less trafficked

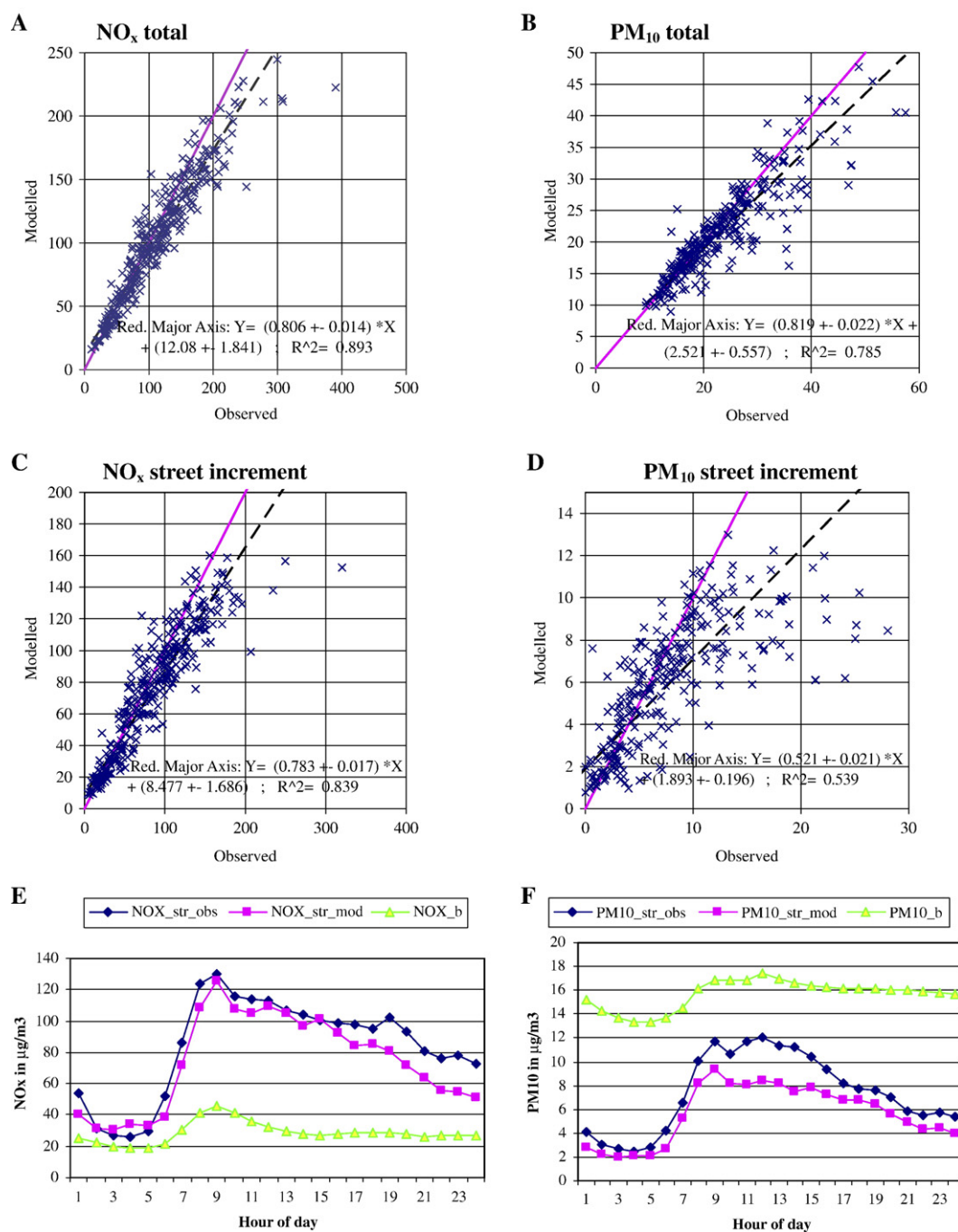


Fig. 1 – Comparison between measurements and OSPM calculations for the monitoring station on Jagtvej in central Copenhagen. The calculations are for the year 2003. Plotted are diurnal mean concentrations (in $\mu\text{g}/\text{m}^3$) for NO_x (A + C) and PM₁₀ (B + D). The upper row (A + B) shows the total concentration while the second row (C + D) shows the results for the street increment only (background concentration subtracted from both modelled and observed). The 1:1 line is shown as solid line. The dashed line gives the best fit to the data using a regression according to the reduced major axis method; fit results given in the figures. Finally is shown the average diurnal variation in observed and modelled street increment and background concentrations for NO_x (E) and PM₁₀ (F).

streets, and the shortest trip using public transport (buses). In the following the “low exposure” route is termed the “green” route, and the “shortest” route is termed the “red” route. Furthermore the public transport route is termed the “bus” route. The morning trip is assumed to have arrival time at the working place at 07:00 hour. Similarly, it is assumed that the trip back from working place to home starts at 14:00 hour. An

additional set of calculations is performed for arrival at work at 09:00 hour and departure from work at 16:00 hour. This second set of calculations is made for determining the impact of travelling inside respectively outside the rush hour time periods in Copenhagen traffic. The calculations are performed for the daily trips over 1 year, where the single individual is commuting to and from work. Subsequently the results are

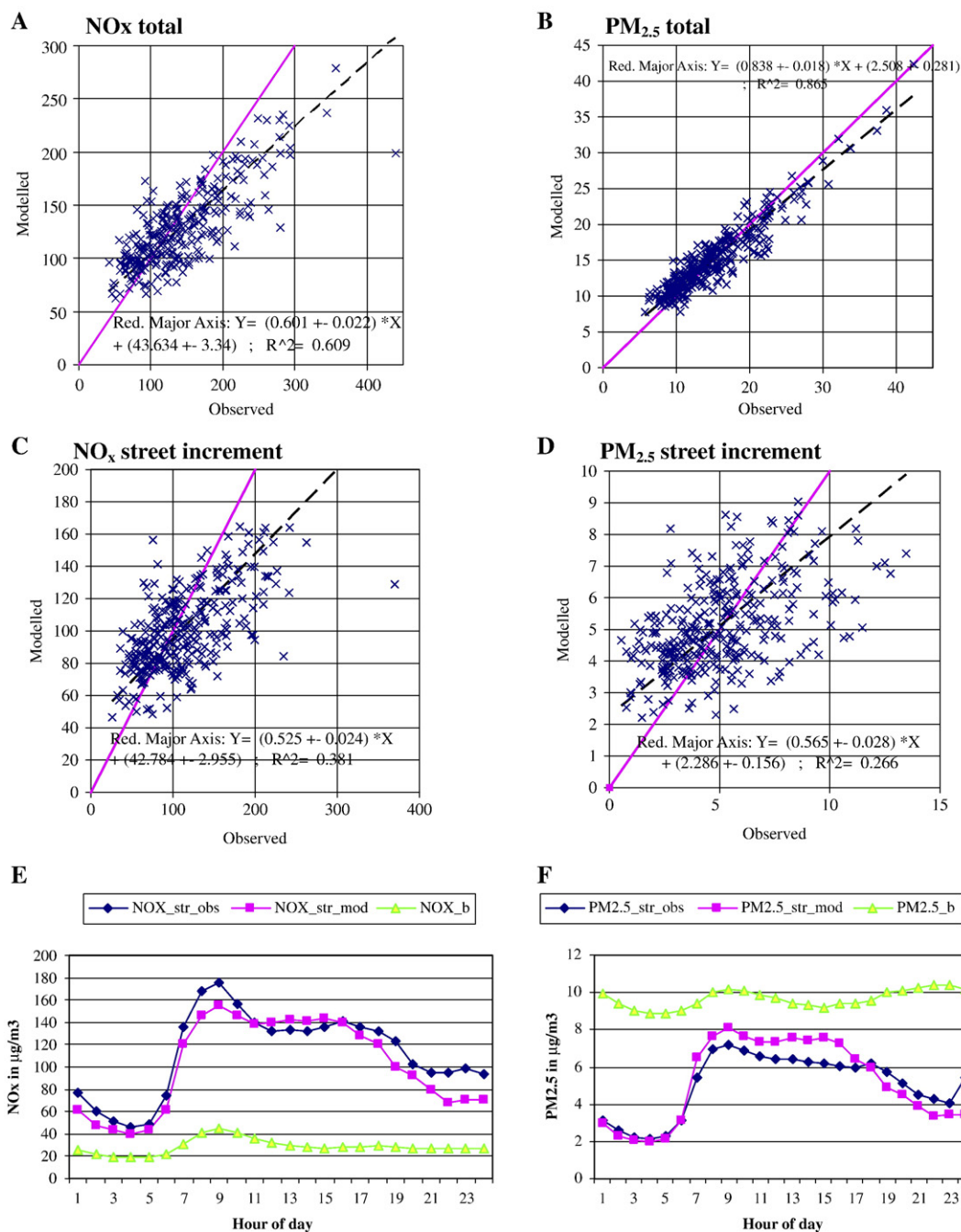


Fig. 2—Graphs as in Fig. 1 for H.C. Andersens Boulevard in 2003, but showing PM_{2.5} instead of PM₁₀.

aggregated. The 1 year calculation period is chosen in order to account for all typical meteorological conditions. A calendar year is assumed to have a total of 250 working days, which is typical in Denmark. Finally it is assumed that all bicycle trips take place with a travel speed of 15 km/h.

The Town Hall in the city centre of Copenhagen is arbitrarily chosen as the working place of the first cohort, and the train station of Ballerup in the sub-urban areas 15 km west of Copenhagen is chosen as the working place of the second cohort. For both cohorts it is assumed that the residences of the 25 selected persons are evenly distributed within a radius of maximum 15 km from the working place (see Fig. 3). The addresses are selected using a random

function, ensuring that the distribution of travel distances for the persons in the cohort comply with the distribution found in a recent analysis of commuting distances in Copenhagen (Danish Bicyclist Association, 2006, personal communication, www.dcf.dk).

A GIS based “route finder”¹ is used to determine the optimal red and green bicycle routes. In order to use this algorithm, each road segment from crossing to crossing is given a cost factor. The red route (the shortest) is here determined simply using distance, i.e. the road segment

¹ ArcGIS 9.1 + NetWork Analyst © by ESRI.

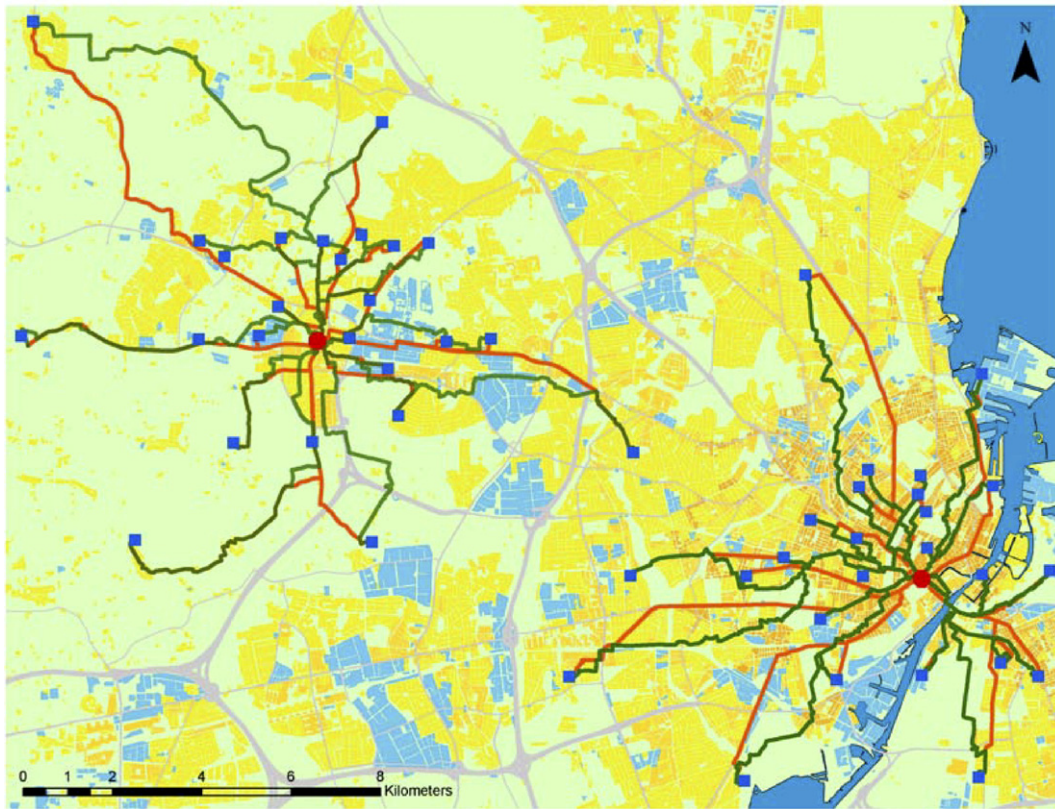


Fig. 3 – Transport routes by bicycle between home and working place for the 2×25 persons in the two selected cohorts. The red lines represent the shortest possible route and the green lines represent the route following less trafficked streets. The home addresses are indicated by blue squares. The routes shown in the left part of the figure are for the sub-urban area Ballerup situated about 15 km from the centre of Copenhagen. The routes in the right part of the figure are going to the Copenhagen city centre. Working places of the two sub-cohorts are assumed to be at the Ballerup train station and at the Copenhagen Town Hall, respectively (these are marked with red dots). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

length, as the cost factor in the analysis. For determining the green route, the Annual Diurnal Traffic (ADT) is taken as a proxy for the air pollution exposure along the segment. Traffic density has in other studies been shown to be a good surrogate index of air pollution exposure (Lipfert et al., 2006). The cost factor for the determination of the green route was established as a combination of distance and ADT. First intuition was to use the product between length of the street segment and ADT. Unfortunately this created unrealistically long D-tours. Travelling few hundred meters along a heavily trafficked street, sometimes became more costly than travelling many kilometres along a less trafficked alternative street. Various alternatives were tested to identify a useful function of distance and ADT that would punish very long routes and create more realistic alternative green routes. On this basis, two functions for the cost factor have been selected and applied for the determination of the green route:

- 1). The third root of the product between ADT and length, and
- 2). The logarithm of the product between ADT and length.

The first cost factor provides the best result in terms of determining the lowest proxy for the total exposure along the

route, but in some cases this cost factor is resulting in an additional travel distance of up to 30% or even 40%. The second cost factor results in route choices with slightly higher proxy for the accumulated exposure along the route, but the additional travel distances (compared with the “red” route) are all below the chosen maximum of 25%. This constrain on the maximum additional travel time was used, since it was considered unrealistic that a person would be willing to travel more than 25% extra before reaching the destination. In reality the route with the lowest accumulated exposure may vary from meteorological situation to another, but this is not accounted for in this investigation.

The Danish Governmental Train Company (DSB – is the Danish abbreviation) operates a web based trip planner (www.rejseplanen.dk). This program (“Rejseplanen” which is Danish for “the travel planner”) has been used to determine the route with the shortest travel time using public bus transport between home and working place. The public transport routes from route maps and route descriptions provided by the travel planner have been digitised manually into the AirGIS system, and subsequently processed in the same way as the bicycle routes. The average travel time for the bus, the red and the green routes where 24 (max 83 min and min 8 min), 14 (max 40 min and min 3 min) and 16 (max 52 min and min 3 min) min, respectively.

4. Obtained results

4.1. Overview

The calculations for the 2×25 fictive persons in the two cohorts provided results for a variety of combinations of pollutants, routes, travel times, concentrations and exposures:

- Pollutants: NO_x , NO_2 , CO, PM_{10} and $\text{PM}_{2.5}$
- Routes: red, green and bus
- Travel times; home to work: 7:00 and 9:00 hour, and work to home 14:00 and 16:00 hour

- Concentration level: total load or street contribution only
- And matrix: concentration/exposure.

The reason for looking at the street contribution separately is that the health effect of pollution from the traffic in the street may be different from the health effect of background pollution; especially when we consider exposure to particle pollution. The exposure is here computed as the sum of products between time spent at a certain segment of the travel route and the corresponding concentration at this segment. The average concentration for a route is defined as the exposure divided by the total travel time for the corresponding route (Hertel et al., 2001a).

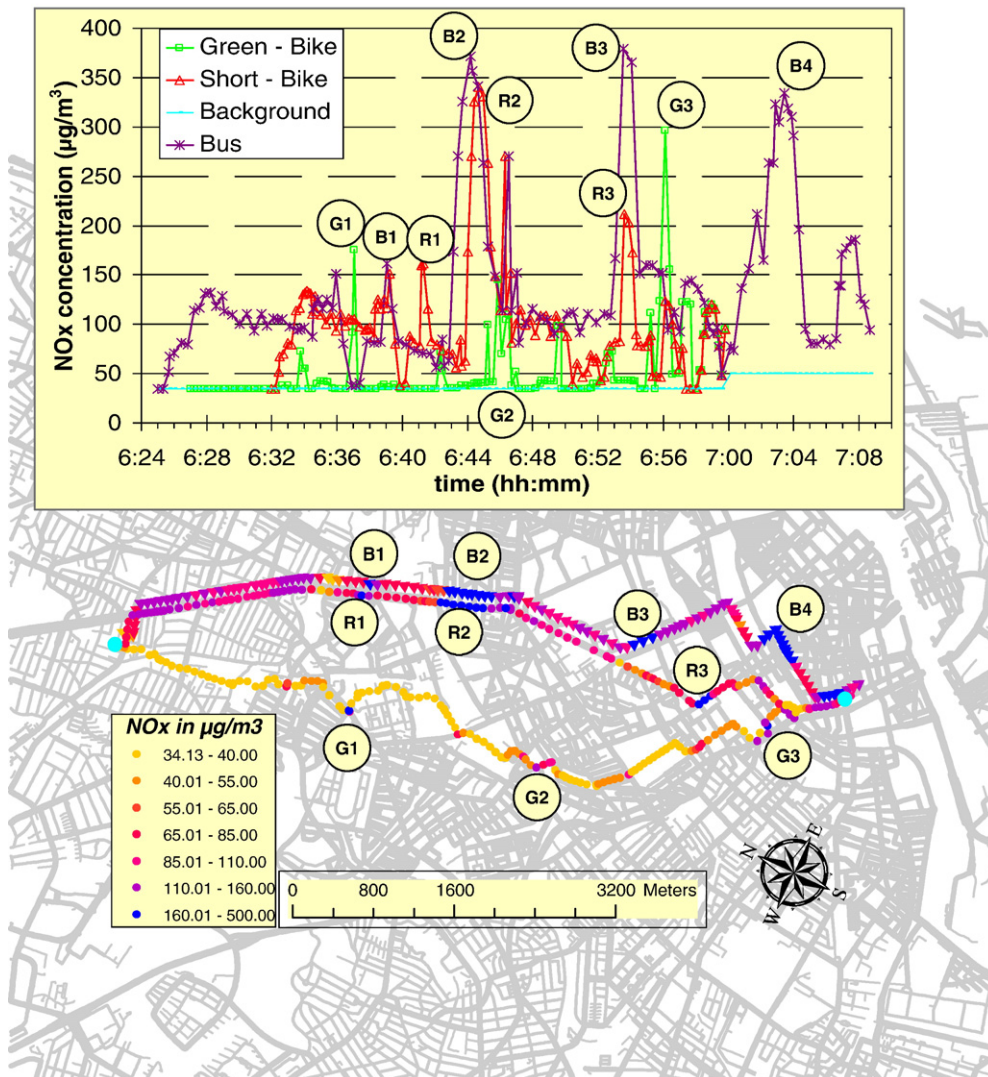


Fig. 4– Average time series of NO_x concentrations (total load) along the three alternative morning (7:00 hour) routes for one specific person travelling to Central Copenhagen. The same data are shown both as a time chart and in their geographic location on the road network. The values are averages for the calculations of the 250 working days and represent therefore the average situation. The background concentration is shown in the time series plot. The width of the road in the under laying road map is related to the daily traffic. Results on the bus route (indicated by triangles) are slightly moved away from the street to avoid overlap with the bicycle routes (circles) in case the travels take place on the same road. The annotated letters R1 to R3, G1 to G3 and B1 to B4 indicate points along the red, green and bus routes where peak values are seen in the time series; peaks that correspond to parts where the route follows or crosses highly trafficked streets.

Out of the large number of combinations of results, a few examples have been selected for illustration of the different levels of aggregation of the results that is discussed in detail later. Fig. 4 shows, as the lowest level of aggregation, the computed concentrations (in this case NO_x total load) for individual street segments for the three alternative morning (07:00) routes for one selected person in central Copenhagen. The figure represents the average concentration along the route for the 250 working days in year (this is the average number of working days in Denmark).

For each of the routes and for each of the different parameters, the average concentrations and average exposures for the 250 working days were calculated. These average results for the 2×25 persons and for the three different routes at four different travel times are plotted for selected examples as scatter plots for accumulated NO_x exposure (Fig. 5) and boxplots (Fig. 6). Finally, the highest level of aggregation in the analysis of the results is shown in Table 1. Here the results for all the 2×25 routes have been averaged for the corresponding combination of the other parameters.

4.2. Results for the sections along single routes

An illustration of the results for NO_x concentrations along the three alternative morning (07:00 hour) routes for one selected person working in Central Copenhagen is shown in Fig. 4. In the figure is shown both time series plot of the obtained concentrations along the routes as well as a geographic presentation of the routes. The two types of plots in the figure are linked by the use of labels. Peak values in concentrations

are evident, when the routes are passing street segments with high traffic density. The presented time series represent the average values for the 250 working days. When single days are considered, the picture may be significantly different due to the impact of the specific meteorological conditions at that particular period of time. In this case the bicycle may pass on the windward side of the street where the pollution concentrations are low, even for a highly trafficked street that may have 5 to 10 times higher concentrations on the opposite (leeward) side of the street.

Fig. 4 shows a typical example with respect to travel times: the red route is shortest and the bus the longest of the routes. The results for the green route are as expected much lower than what is obtained for the other route. Often the results for the green route are close to the background levels. In other cases, the green route is less optimal and passes highly trafficked street segments.

4.3. Results averaged for single routes

Fig. 5 shows the results for the accumulated NO_x exposures along the 2×25 routes, where the green and the bus routes are plotted versus the red routes. The accumulated air pollution exposure shows in average a significant difference between the green and the red routes. Still there are a few study subjects that receive a smaller accumulated exposure along the red route compared with the green route, even when the primary pollutants are considered. Again this is the result of the difference in travel time along the two routes.

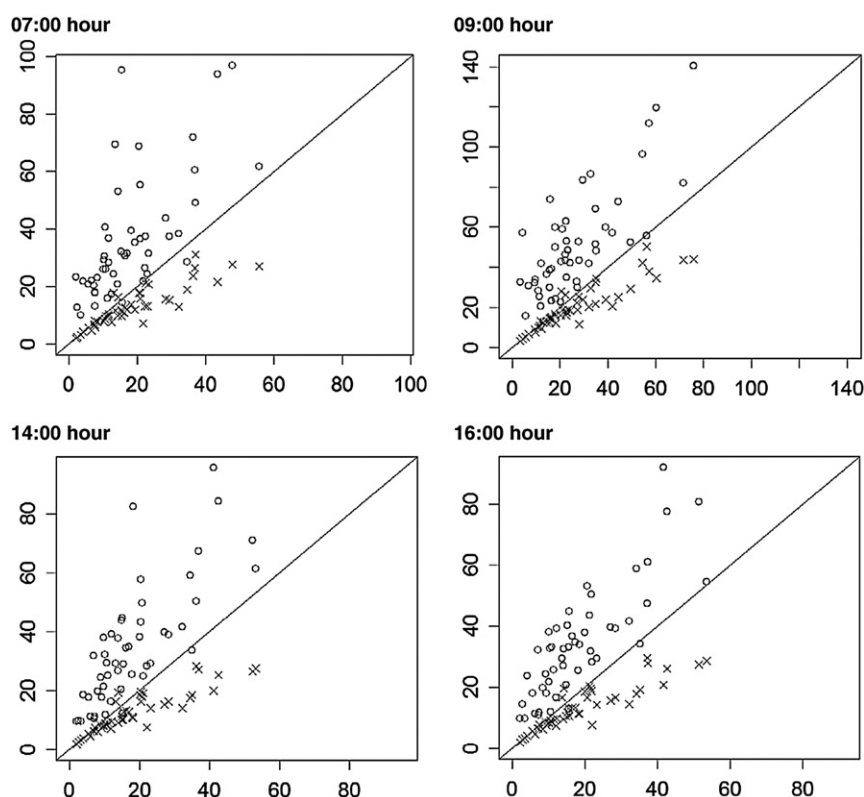


Fig. 5 – Plot of NO_x exposures ($\mu\text{g}/\text{m}^3 \times \text{h}$) for the green (x) and the bus (o) routes versus the red routes. Average values for the 50 routes are shown in separate plots for the travel times 07:00, 09:00, 14:00 and 16:00 hour.

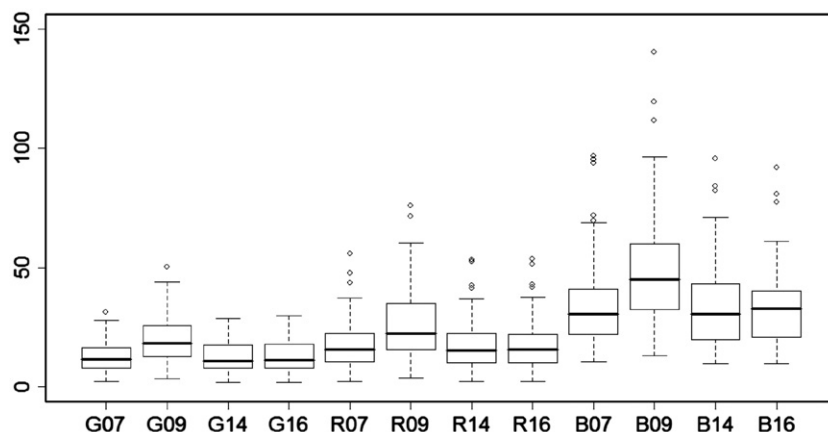


Fig. 6–Boxplot for the accumulated NO_x exposure ((μg/m³) × h). Data are shown for the green (G), red (R) and bus (B) routes for the four different travel times (07:00, 09:00, 14:00, and 16:00 hour). The bold line in the box indicates the median value. The upper and lower borders of the box indicate the 75 and 25 percentile values, respectively. The Whiskers indicate the range for the 95% confidence interval, and finally are outliers indicated by small circles. The plot is generated using R software (R Development Core Team, 2006).

4.4. Average concentrations for all routes

The all-route average air pollution concentrations are in general between 10% and 39% lower along the green route compared with the red route (Table 1). The difference is considerably larger (between 54% and 67%) when considering the street contribution – the contribution from traffic in the considered street and disregarding the urban background contribution. The street contribution is here obtained for each street segment by subtracting the urban background level from the total street pollution concentration. For the secondary pollutants – PM₁₀, PM_{2.5} and NO₂ – the difference between the red and the green route is only moderate (between 10% and 14%) when considering the average pollutant concentrations along the route. However, the difference between the red and the green route is more or less similar (between 54% and 66%) to those obtained for the primary pollutants (NO_x and CO) when the street contribution is considered.

4.5. Average exposure

The travel time is in average about 15% longer for the green route compared with the red route. This additional travel time means that the difference between the green route and the red route in accumulated air pollution exposure is insignificant or in many cases even negative for the secondary pollutants. For accumulated exposure to CO, the difference is between 12% and 19%; and for NO_x between 24% and 31%. The accumulated exposure to the street contribution shows a significant difference for all pollutants of between 54% and 66%. These results are well in line with the results obtained for the average concentrations along the routes.

The average pollutant concentration along the route is for all pollutants lower when choosing the green route instead of the red route (Table 1).

4.6. Secondary versus primary pollutants and influence of background

The secondary pollutants – PM₁₀, PM_{2.5} and NO₂ – in urban streets are to a large extent dominated by the contribution from urban background concentrations. The specific model calculations point at a contribution in the range of 70% to 80% of the average concentration along the route. For the mass of particulate matter (PM₁₀ and PM_{2.5}), this is mainly a result of the long-range transported air pollution. For NO₂, the concentration is to a large extent dominated by chemical conversion of NO in the reaction with long-range transported O₃. The large contribution from urban background means that little difference in average air pollution concentration is obtained between choosing the red and the green routes. Since the travel time of the green route is longer than the travel time of the red route, this means as previously mentioned that the difference in accumulated exposure along the routes is very small and in most cases even negative for the secondary pollutants.

4.7. Choice of route summary

For the “bus” route the average concentration along the route is between 2% and 15% higher than the results obtained for the red route. For the accumulated exposures along the bus route, the difference is even more pronounced with between 75% and 115% higher values for the bus route compared with the red bicycle route. The bus follows in general highly trafficked streets. In almost all cases the bus gives higher average concentrations and accumulated exposures than the red routes, whereas the green routes always give lower average concentrations and in far the most of the cases also lower exposures.

4.8. Choice of travel time

The results obtained for the average air pollution concentration as well as accumulated pollutant exposure along the

Table 1 – Average pollutant concentration ($\mu\text{g}/\text{m}^3$) and pollutant exposure ($(\mu\text{g}/\text{m}^3) \times \text{hour}$)

Pollutant	Time	Bicycle route and total load						Bicycle route and street contribution						Bus route and total load				Bus route and street contrib			
		Concentration			Exposure			Concentration			Exposure			Conc.		Exposure		Conc.		Exposure	
		Red	Green	Diff.	Red	Green	Diff	Red	Green	Diff	Red	Green	Diff	Bus	Diff	Bus	Diff	Bus	Diff	Bus	Diff
NO _x	7	79.3	49.2	38	18.14	12.68	30	45.1	15.1	67	10.25	3.44	66	90.3	–14	36.30	–100	55.9	–24	22.04	–115
	14	79.6	48.9	39	18.20	12.50	31	48.7	18.0	63	11.05	4.14	63	91.5	–15	35.01	–92	59.9	–23	22.35	–102
	9	115.9	78.1	33	26.55	20.16	24	60.6	22.8	62	13.75	5.19	62	126.9	–9	51.00	–92	72.4	–19	28.40	–107
	16	80.7	50.5	37	18.43	12.93	30	48.3	18.2	62	10.95	4.17	62	88.7	–10	33.91	–84	56.4	–17	20.98	–92
NO ₂	7	32.7	28.8	12	7.52	7.68	–2	6.8	2.9	57	1.52	0.66	57	34.4	–5	14.05	–87	8.4	–24	3.26	–114
	14	30.6	26.2	14	6.99	6.94	1	7.9	3.5	56	1.73	0.79	54	33.3	–9	12.96	–85	9.7	–23	3.49	–102
	9	41.7	37.5	10	9.57	10.00	–4	7.7	3.5	55	1.71	0.80	53	42.9	–3	17.55	–83	9.3	–21	3.62	–112
	16	32.7	28.4	13	7.48	7.53	–1	7.9	3.6	54	1.74	0.82	53	34.1	–4	13.31	–78	9.3	–18	3.36	–93
CO	7	455.6	351.2	23	104.38	92.01	12	173.0	68.5	60	38.98	15.49	60	513.2	–13	206.89	–98	228.7	–32	88.72	–128
	14	560.2	410.1	27	127.91	106.66	17	252.0	101.9	60	56.59	23.22	59	609.2	–9	245.87	–92	298.1	–18	114.78	–103
	9	758.2	570.4	25	173.02	148.19	14	323.8	136.0	58	72.51	30.59	58	815.4	–8	328.54	–90	386.0	–19	150.87	–108
	16	694.9	497.1	28	158.34	128.70	19	333.6	135.8	59	74.75	30.91	59	755.1	–9	289.45	–83	394.8	–18	145.11	–94
PM ₁₀	7	18.0	15.9	12	4.16	4.26	–2	3.2	1.1	66	0.74	0.26	65	18.6	–3	7.65	–84	3.8	–19	1.49	–101
	14	20.8	18.5	11	4.79	4.95	–3	3.7	1.4	62	0.83	0.32	61	21.2	–2	8.39	–75	4.4	–19	1.63	–96
	9	22.6	19.8	12	5.20	5.29	–2	4.4	1.7	61	1.01	0.38	62	23.1	–2	9.45	–82	5.1	–16	1.99	–97
	16	20.0	17.8	11	4.61	4.75	–3	3.6	1.4	61	0.81	0.32	60	20.4	–2	8.08	–75	4.0	–11	1.50	–85
PM _{2.5}	7	11.6	10.1	13	2.68	2.70	–1	2.3	0.8	65	0.53	0.18	66	11.9	–3	4.91	–83	2.6	–13	1.04	–96
	14	11.7	10.2	13	2.69	2.71	–1	2.4	0.9	63	0.55	0.21	62	12.3	–5	4.83	–80	2.8	–17	1.05	–91
	9	13.5	11.6	14	3.10	3.08	1	3.0	1.1	63	0.69	0.26	62	13.8	–2	5.62	–81	3.4	–13	1.33	–93
	16	11.7	10.2	13	2.69	2.73	–1	2.3	0.9	61	0.53	0.20	62	11.9	–2	4.70	–75	2.5	–9	0.95	–79

Calculations are performed for working days over 1 year for the travel along the bicycle and bus routes. Travels for inside (7 and 14) and outside (9 and 16) rush hours in Copenhagen are presented in the table. For the bicycle route, the results for the Red and the Green routes are presented together with the difference between Red and Green routes given in percent. Positive difference means less concentration/exposure for the green route. Negative difference means higher concentration for the green route/bus route compared with the red route. For both Bicycle and Bus routes, the results are furthermore shown both for the total load (street generated plus background) and for streets contribution (only generated in the street). For the bus route, only one route is analysed and results are again presented for the total load and for the street contribution (as well as for the four different travel times). Presented together with the results for the Bus route are also the difference in percent between Bus route and Red route. For all the calculated difference a reduction in concentration or exposure is given as a positive value and an increase as a negative value.

routes show a significant effect of travelling within respectively outside the time window of the rush hours in the Copenhagen traffic. Again the effect is somewhat stronger for the primary (10% to 30%) compared with the secondary (5% to 20%) pollutants (see Table 1).

The difference between rush hour and non rush hour time periods is even more clear when looking at the results in a boxplot. Such plots are shown for NO_x exposures in Fig. 6. The impact of the rush hour on concentrations and exposures is pronounced especially for the morning hours, but the effect is much less evident for the afternoon hours. This is expected since the traffic peak in the afternoon rush hour time period is stretched over several hours compared with the relatively short and sharp morning rush hour. The effect is furthermore most evident for the bus routes that in most cases follow highly trafficked streets.

5. Discussion

This section contains a discussion of the possible shortcomings and limitations in the chosen methodology.

The OSPM computes hourly mean air pollution concentrations for the various street segments along the routes. For a specific travel, the residence time for a given study subject within a specific street segment will be in the order of seconds. When considering a time scale of seconds, the pollutant concentrations will in some cases reach peak values that are considerably higher than what is computed by the model. However, the model calculations are considered to represent reasonably well the average conditions over the 250 working days.

The analysis of bicycle travels inside and outside the rush hour time periods is based on a fixed travel speed. In reality there may be street segments where the traffic stops or slow down considerably during rush hours. Bicycles will in some cases be able to continue the travel, despite the cars in the street are moving slowly. Still it must be expected that the travel through the busiest streets will take longer during rush hours, and this is not accounted for in the presented calculations. In rush hour situations where the traffic in the street slow down considerably or even stops, this is in some cases also affecting the turbulence in the street. During low wind speed conditions, the traffic induced turbulence is significant for the dispersion of traffic pollutants in the street (Berkowicz et al., 1996). This effect is not taken into account in the performed calculations. Both these effects will therefore lead to somewhat more pronounced differences between green and red routes.

The urban background concentrations are for all the presented calculations obtained from measurements performed at the monitoring station on the H.C. Ørsted Institute in central Copenhagen. For the routes to and from Ballerup station, these urban background levels will in reality be smaller, since Ballerup is a sub-urban area outside Copenhagen. This simplification concerns the total average concentration and the total accumulated exposure along the routes. The street contribution is unaffected of this simplification since the urban background in this case is disregarded. A smaller urban background would result in a somewhat larger difference between red and green route for the Ballerup area.

The determination of the green route was carried out using diurnal traffic as a proxy for the air pollution exposure. The optimal way of determining the green route would be to compute the air pollution concentrations for each street segment in the entire road network, and use these values directly as the cost factor in the path finder routine. For example can a street with lower daily traffic but higher or narrower surrounding buildings have higher air pollution than a more trafficked street with more open surroundings; i.e. better dilution conditions. The analysis of the results have shown examples (not plotted in this paper) where the green route contains several sections with relatively high pollutant concentrations compared with the red route. A more correct procedure using mapping of air pollution concentrations for an entire network of streets for finding the best route would, however, be highly demanding in terms of computer resources. Furthermore, it is not considered likely that substantially different results would have been obtained, but this should be validated in future studies. The actual meteorological situation governs the actual air pollution level in the streets including the difference in pollution level between the two sides of the street. Therefore, there might still be some situations where a slightly different route would be chosen in case hour by hour pollutant concentrations had formed the basis for the selection of route.

Concerning the results for the bus route, it should be noted that the travel from home and working place to the nearest bus stop has not been accounted for in the calculations. This trip may in some cases be of 5 to 15 min walk, and will in most cases be along streets with relatively low air pollution levels. This means that the average pollutant concentration along the bus route is somewhat overestimated for some of the routes. However, the travel time is at the same time underestimated in these cases which will pull the results for the accumulated exposure along the route in the opposite direction. It is believed, however, that this simplification does not affect the results significantly.

6. Conclusions and perspectives

Concluding from our presented investigations, we can provide the following answers to the questions raised at the end of the introduction:

The presented study has demonstrated that it is possible to significantly reduce the accumulated air pollution exposure during the daily bicycle route between home and working place by following the low exposure route instead of following the shortest possible route. The difference in accumulated air pollution exposure is significant for the primary pollutants. When the street contribution is considered the difference in accumulated air pollution exposure is significant for both primary and secondary pollutants.

Travelling outside the rush hour time periods significantly reduces the accumulated air pollution exposure along the routes through the city. The effect is seen to be especially pronounced for the morning rush hours, whereas the effect is less pronounced for the rush hour in afternoon. The population may therefore be advised to travel outside the rush hour time periods.

The urban background is very important for the exposure to secondary air pollutants. This means that for the secondary pollutants, little difference is seen in average concentrations along the routes. For the accumulated air pollution exposure to secondary pollutants the effect of following the green route is insignificant and in many cases even negative. This is a result of the little concentration difference in combination with the longer travel time for the green route compared with the red route.

The study has shown that the accumulated pollutant exposure is much higher for the bus route compared with the red route. The bus follows highly trafficked streets and in addition the travel time is longer since it usually does not follow the most direct route to the destination.

The obtained results from this study indicate that there is a potential for developing a route planner to help the population in choosing the low exposure routes through the city. Such a route planner would make an alternative to the currently available route planners for choosing the route with the shortest distance or the shortest travel time.

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